Burdick Crag Mass: A Substrate Wave Model for Galactic Rotation Curves Without Dark Matter

v7 — Anchor Partition Regime Map, ROOT_REENTRY Anatomy, and Macro-Torsion Operator

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Correspondence: GitHub: Joy4joy4all/Burdick-Crag-Mass **Framework DOI:** 10.5281/zenodo.19251192 **Paper version:** v6 (2026-04-21) **Data provenance:** Full 175-galaxy SPARC analysis from BCM_v26_Tully_Fisher_Amplitude_Scaling_Ablation (Test 2 rev 6, completed 2026-04-21) and bracket-stratified Tully-Fisher bootstrap from BCM_v26_Tully_Fisher_Exponent_Stability (Test 4, completed 2026-04-21).

Version History

v7.0 (2026-05-30): Anchor Partition Ratio (APR) decomposition across all 175 SPARC galaxies. Six-regime structure identified. 125 km/s snap confirmed (delta=+0.193). ROOT_REENTRY anatomy (6 galaxies, v_max > 300 km/s). Macro-Torsion operator proposed, tested regime-safe (5/5 gates, Sections 3.8-3.9). Volume Dilatant rejected as formulated. Conclusions updated. All v6 content retained verbatim.

v6.0 (2026-04-21): Full SPARC 175 analysis, amplitude ablation, Tully-Fisher exponent stability (alpha=3.42, bracket breakdown at massive v_max).

v5.0 (2026-04-15): Einstein coupling, null pump proof, boundary dynamics, σ_crit saturation limit.

v1–v4: Progressive 29-galaxy to full SPARC analysis, Tully-Fisher recovery, neutrino flux predictions.

Abstract

We present the Burdick Crag Mass (BCM) framework, a substrate-based phenomenological alternative to cold dark matter in which the dark matter signal is modeled as the energy maintenance budget of a pre-existing 2D spatial substrate. The substrate field becomes gravitationally significant only when continuously agitated by wave energy; we propose supermassive black hole neutrino flux as the physical funding mechanism. Using four global parameters (lambda=0.1, kappa=2.0, grid=256, layers=8) plus one amplitude coefficient fit per galaxy, we evaluate the framework on the complete 175-galaxy SPARC rotation curve sample (Lelli et al. 2016). BCM beats Newtonian baryonic fits on 130 of 175 galaxies (74.3%) and beats MOND on 172 of 175 (98.3%), with mean reduced chi-squared of 63.1 compared to Newton's 157.6 and MOND's 1122.6. An ablation study confirms that a single global amplitude cannot replace per-galaxy fitting: the 1-parameter-per-galaxy rescaling outperforms a 1-parameter-total global fit by a factor of 13.8. The per-galaxy amplitude coefficient remains a single scalar with no free shape parameters, making BCM competitive with NFW dark matter halo fits (2 parameters per galaxy) and MOND (global acceleration scale). Bracket-stratified analysis reveals the framework classifies galaxies into physically distinct substrate interaction states, with BCM dominating low-to-mid velocity regimes (88% wins in v_max=50-100 km/s, 82% in 100-150 km/s) and showing systematic weakness in the massive bracket (v_max > 200 km/s, 61% wins). The Tully-Fisher exponent on the full sample is alpha = 3.42 (95% CI [3.03, 3.88], consistent with classical baryonic TF), but bracket-stratified bootstrap reveals substantial scale-dependence: the TF relation breaks down entirely at the massive bracket (alpha = -0.29, CI fully below classical), suggesting a scale-dependent phase transition in the substrate coupling that will be explored in a forthcoming companion paper. The framework is formulated as a classical field theory with a well-defined action and Lagrangian density. We extend the framework to binary stellar dynamics across three systems (Alpha Centauri, HR 1099, Spica) and reproduce the structural phases of the GW150914 binary black hole merger. Falsifiable predictions are provided for IceCube/KM3NeT neutrino flavor ratio gradients at galactic edges, Betelgeuse m=3 tachocline recovery (circa 2030-2031), and binary mass transfer rates in synchronized vs unsynchronized systems. All code, data, and timestamped results are publicly available across 26 versioned Zenodo publications with full adversarial audit records from four independent AI systems.

Version 7.0 additions: The Anchor Partition Ratio (APR) is applied to all 175 SPARC galaxies as a direct decomposition of observed rotation curves, requiring no BCM solver execution. APR identifies six empirical substrate regimes; the dominant structure is a substrate plateau at 50-150 km/s (APR_outer median 0.77, BCM wins 91.9%) and a suppression valley at 150-300 km/s (APR_outer median 0.47, Newton wins 75.7%). A sharp APR discontinuity at 125 km/s (delta = +0.193) aligns independently with the Tully-Fisher bracket transition reported in v6. Six ROOT_REENTRY galaxies (vmax > 300 km/s) show high outer-disk APR but BCM underfit; anatomical analysis confirms outer-disk substrate presence in 5/6. A Macro-Torsion operator $T = \eta \cdot \Theta(v_{max} - 300\ \text{km/s}) \cdot |\partial v_\phi/\partial r - v_\phi/r|$ is introduced, tested against all six APR regimes, and confirmed structurally regime-safe: it produces zero effect on all galaxies below 300 km/s (0.000% η sensitivity, 5/5 gates) while recovering ROOT_REENTRY outer RMS from 113.5 to 58.4 at test parameters. The operator is not yet integrated into the BCM solver; it is a confirmed directional extension.

Keywords: dark matter alternative, rotation curves, SPARC, substrate wave model, neutrino flux, galactic classification, binary dynamics, phase coherence, anchor partition ratio, regime map, ROOT_REENTRY, macro-torsion operator, substrate plateau, suppression valley

1. Introduction

The dark matter problem remains one of the central unsolved questions in astrophysics. Observed galactic rotation curves (Rubin & Ford 1970; Bosma 1981) require either substantial invisible mass or modifications to gravitational theory. The standard Lambda-CDM cosmological model postulates cold dark matter halos surrounding galaxies, reproducing large-scale structure but encountering persistent difficulties at galactic scales: the cusp-core problem (de Blok 2010), the missing satellites problem (Klypin et al. 1999), and the too-big-to-fail problem (Boylan-Kolchin et al. 2011). Direct detection experiments have found no particle candidates despite decades of effort.

Modified gravity approaches, particularly MOND (Milgrom 1983) and its relativistic extensions (Bekenstein 2004), successfully reproduce individual galaxy rotation curves but require per-galaxy fitting of the mass-to-light ratio and encounter difficulties with galaxy clusters and cosmological observations.

We propose a third approach: the dark matter signal is neither missing mass nor modified gravity but the observable signature of a substrate maintenance cost. A pre-existing 2D substrate medium becomes gravitationally significant only when continuously agitated by wave energy, analogous to how a vibrating drumhead reveals its geometry through standing wave patterns. The proposed funding source is the supermassive black hole at the galactic center, continuously injecting energy into the substrate via neutrino flux. Where the funding stops, the substrate decays and the gravitational signal disappears — producing the rotation curve profiles attributed to dark matter halos.

The framework is formulated as a classical field theory with a well-defined action and Lagrangian density (Section 2.5), from which the equations of motion are derived via the Euler-Lagrange principle.

This paper presents the mathematical framework (Section 2), the 29-galaxy chi-squared comparison against NFW and MOND (Section 3), binary stellar dynamics on three systems (Section 4), falsifiable predictions (Section 5), and discussion of limitations and future work (Section 6).

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2. The Substrate Model

2.1 Wave Equation

The substrate field sigma evolves on a 2D grid according to:

```
d^2 sigma / dt^2 = c^2 * laplacian(sigma) - lambda * sigma
                   + S(x,t) + alpha * (sigma - sigma_prev)
                   - gamma * d_sigma/dt
```

where sigma is the substrate memory field (cumulative agitation), c is the wave speed, lambda is the decay rate (the maintenance cost — higher lambda means faster decay without continuous funding), S(x,t) is the SMBH neutrino flux injection source, alpha = 0.80 is the memory coefficient governing how much substrate state persists between timesteps, and gamma is the damping coefficient.

The memory term alpha*(sigma - sigma_prev) is critical: it provides the substrate with history-dependent behavior. At alpha = 0.80 the system operates in a stable regime; a sharp bifurcation occurs at alpha = 0.90 (v14 Lagrange equilibrium scan), establishing the operating window at 0.75 to 0.85.

2.2 Crag Mass Injection

The SMBH source term uses Bessel function modes:

```
J_crag = kappa_BH * (M_BH / M_ref) * J_m(k * r / r_max)
```

where kappa_BH = 2.0 is frozen globally (never tuned per galaxy), J_m is the Bessel function of the first kind of order m, k is the wavenumber set by the tachocline gate, and r is normalized to the grid radius r_max = grid/2. The mode m is selected by the tachocline gate based on the stellar/BH rotation profile. This produces concentric ring patterns in the substrate field that map to the observed rotation curve structure.

Black hole masses are obtained from the catalog of Kormendy & Ho (2013) where available, or estimated from the M_BH-sigma relation of Tremaine et al. (2002) using V_max / sqrt(2) as a velocity dispersion proxy.

2.3 Rotation Curve Comparison

The substrate-modified circular velocity is:

```
v_sub(r) = sqrt(v_newton^2(r) + kappa * sigma(r))
```

where v_newton(r) is computed from baryonic components (gas + disk + bulge) provided by SPARC Rotmod files (Lelli et al. 2016).

The substrate velocity contribution is extracted from the solver's gravitational potential via the clean chain:

```
rho_eff = rho^2        (energy density of agitation field)
Psi = Poisson(rho_eff) (gravitational potential)
v_sub = sqrt(r * |dPsi/dr|)
```

The substrate contribution replaces the NFW dark matter halo term. Per-galaxy amplitude scaling is applied: the substrate velocity profile shape is the test quantity, not the absolute amplitude.

2.4 Global Parameters

Four parameters are fixed globally across all 175 galaxies:

Parameter	Symbol	Value	Role
Decay rate	lambda	0.1	Substrate maintenance cost
BH coupling	kappa_BH	2.0	SMBH injection strength
Grid resolution	grid	256	Spatial resolution
Entangled layers	layers	8	Substrate depth

No per-galaxy tuning occurs. The six-class taxonomy (Section 3) emerges from the interaction between global parameters and each galaxy's specific baryonic mass distribution and SMBH properties.

2.5 Action and Lagrangian Density

The substrate model is derived from a variational principle. The action is:

```
S = integral [ L ] d^2x dt
```

with Lagrangian density:

```
L = (1/2)(d_mu sigma)(d^mu sigma)
    - (lambda/2) sigma^2
    + J_crag * sigma
    + (alpha/2)(sigma - sigma_prev)^2
```

The first term is the kinetic energy of wave propagation. The second is the mass term encoding the maintenance cost. The third couples the SMBH neutrino flux to the substrate. The fourth is the memory term, which in the continuous limit (sigma - sigma_prev)/dt approaches a first-order time derivative coupling, standard in dissipative field theories for open thermodynamic systems.

Variation with respect to sigma yields the Euler-Lagrange equation:

```
d^2 sigma/dt^2 = c^2 laplacian(sigma) - lambda*sigma
                 + J_crag + alpha*(sigma - sigma_prev)
```

which reproduces the solver's wave equation (Section 2.1).

The stress-energy tensor is defined as:

```
T_mu_nu = d_mu(sigma) * d_nu(sigma) - g_mu_nu * L
```

The substrate memory field sigma sources an effective gravitational contribution through this tensor. The connection between the 2+1 dimensional substrate field and 3+1 dimensional spacetime curvature is the subject of ongoing work (see Section 6.2).

2.6 Numerical Implementation

The wave equation is discretized on a uniform 2D Cartesian grid using the following scheme:

Time integration: Störmer-Verlet (leapfrog):

```
rho^(n+1) = 2*rho^n - rho^(n-1)
            + dt^2 * (c^2 * lap(rho^n) - lambda*rho^n + J)
            - gamma * dt * (rho^n - rho^(n-1))
```

Spatial discretization: Five-point Laplacian stencil:

```
lap(f)_ij = (f_{i+1,j} + f_{i-1,j} + f_{i,j+1} + f_{i,j-1}
             - 4*f_{ij}) / dx^2
```

Boundary conditions: Absorbing layer at grid edges. An exponential taper over 10 grid cells damps the field to zero at all four boundaries, preventing spurious reflections.

Inter-layer coupling: Eight substrate layers are coupled via nearest-neighbor entanglement with coupling coefficient epsilon = 0.02:

```
transfer = epsilon * (field_l - field_{l+1})
```

CFL condition: c * dt / dx < 1.0 is enforced at initialization.

Convergence protocol: Each galaxy is evolved for 20,000 settle steps followed by 5,000 measurement steps. The measurement-phase average of rho and sigma provides the converged field. The settle period eliminates transient initialization artifacts.

Poisson solver: Jacobi iterative relaxation with 20,000 iterations and tolerance 1e-7, using Dirichlet boundary conditions (phi = 0 at all edges).

Computational cost: The production run on all 175 SPARC galaxies at grid=256 with 25,000 integration steps per galaxy required 148.6 minutes wall time on a single NVIDIA GPU (Windows 10/11, Python 3.11, CuPy, Anaconda terminal).

Grid convergence: Six galaxies change winner classification from non-BCM to BCM when resolution increases from grid=128 to grid=256, consistent with the resolution boundary documented in v14 of the framework development record. Grid=256 is used for all reported results.

2.7 The Anchor Equation

The substrate model is grounded in a single governing equation — the Anchor Equation — which unifies the classical mass-energy relation with the substrate corridor contribution. This equation has been developed progressively from v26 through v30 of the framework; its full form is stated here for the first time in Paper A.

Two-term form (v26 baseline):

```
E = M·Phi(sigma)·c²  +  ■_{Aleph-0} J·dl
```

The first term, M·Phi(sigma)·c^2, recovers Einstein's $E = mc^2$ as sigma approaches zero (Phi → 1). This is the Einstein recovery limit, verified numerically at Alpha Centauri (Phi deviation 3.35e-10, three to eight orders of magnitude below the 5% falsification threshold).

The second term, the closed contour integral of substrate current J around a topological Aleph-Null loop, activates only when the substrate field is driven toward sigma_crit by paired pump sources (supermassive black holes). At a single isolated pump or in the absence of substrate (sigma → 0), J = 0 and this term vanishes identically.

The coupling efficiency Phi(sigma) is:

```
Phi(sigma) = 1 - tanh²(sigma / sigma_crit)
```

Properties: Phi(0) = 1 (empty substrate, pure Einstein); Phi(sigma_crit) → 0 (saturated substrate, full corridor regime); dPhi/dsigma < 0 monotonically decreasing.

Extended five-term form (v27):

```
E = M·Phi(sigma)·c²
    + ■_{Aleph-0} J·dl
    + integral[ P_{7D→3D} / R_{9→10} ] dXi
    + R(nu · grad(G))
    ± T(Psi_tach)
```

Term 3 integrates the 7D-to-3D projection operator P_{7D→3D}(OpT, OpC) across the latent substrate structure Xi, normalized by the R_{9→10} gate functional. This term governs how much substrate information survives the dimensional collapse from the full 9D ontological stack to the observable 3D+time. When the gate closes ($R_{9\to10} \to 0$), the integrand becomes large and integration terminates — the gate closure physically represents a substrate coherence threshold.

Term 4, R(nu · grad(G)), implements post-Poisson snap-back. It operates on the substrate velocity field nu dotted with the gradient of the field-shape generator G. This term produces the tare-edge behavior observed at the Brucetron Hemorrhage threshold in v17-v19 tests, and is responsible for boundary nonlinear behavior at sigma_crit saturation.

Term 5, ±T(Psi_tach), represents a forward-lead tachyon contribution with sign ambiguity. The functional form is deferred pending operational coupling specification. It is not incorporated in any current solver computation.

Recovery property: at sigma → 0, all terms except Term 1 vanish identically, and $E \to M\cdot c^2$. The full equation reduces to Einstein in the classical limit. This recovery is enforced as a falsification constraint: any test run that violates it is rejected as solver error.

Connection to the Anchor Partition Ratio (v30):

The APR decomposition introduced in Section 3.8 is the observational projection of the Anchor Equation onto the measurable velocity excess of SPARC rotation curves:

```
APR(r) = substrate-side term² / (mass-side term² + substrate-side term²)
```

At APR → 0: the mass term M·Phi(sigma)·c² dominates; Newton explains the curve. At APR → 1: the corridor term ■ J·dl dominates; substrate accounts for most of the velocity excess. The six-regime structure (Section 3.8) maps which galaxies sit in which regime of the Anchor Equation's two-term balance.

Connection to the Macro-Torsion Operator (v30):

The Macro-Torsion operator proposed in Section 3.9 extends the right-hand side of the Anchor Equation for ROOT_REENTRY systems by adding a velocity-shear torsion tensor:

```
T_munu = eta · Theta(vmax - 300 km/s) · |dv_phi/dr - v_phi/r|
```

This is a proposed additional term to the substrate corridor side of the Anchor Equation, gated by the Heaviside function to activate only when the galactic rotation velocity exceeds 300 km/s — the ROOT_REENTRY threshold identified in Section 3.7. At this threshold, the standard two-term Anchor Equation underestimates the outer-disk substrate contribution, and the torsion term provides the missing coupling.

Calibration and open questions:

The Anchor Equation as formulated is phenomenological. The connection between the 2+1D substrate field and the 3+1D stress-energy tensor is developed in Section 2.5 but not yet closed into a derivation from first principles. Term 5 (tachyon contribution) has not been assigned an operational form. The $R_{9\to10}$ gate functional is operationally defined through the OpT/OpC proxy values but not yet anchored to a physical observable. These are the primary theoretical targets for Papers B and C.

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3. SPARC Rotation Curve Classification

3.1 Dataset

We use the complete 175-galaxy SPARC dataset (Lelli et al. 2016), which provides observed rotation curves and baryonic mass models (gas + disk + bulge) for late-type galaxies spanning five velocity brackets: dwarf (V < 50 km/s), low (50-100), mid (100-150), high (150-200), and massive (V > 200 km/s).

3.2 Six-Class Substrate Taxonomy

Running the substrate solver on all 175 galaxies with fixed global parameters produces a natural classification into six physically distinct substrate interaction states:

Class I — Transport-Dominated: The substrate is the dominant contributor to the rotation curve. The SMBH maintains a strong, well-funded torus extending beyond the visible disk. Substrate RMS wins over Newton by significant margins (e.g., NGC 2841, +28.4 km/s).

Class II — Residual/Hysteresis: Substrate contribution is present but secondary. The galaxy is in a transitional state between full substrate dominance and ground state.

Class III — Ground State: Minimal substrate activity. The baryonic mass distribution alone nearly explains the rotation curve. The SMBH injection is weak or the galaxy is too diffuse to maintain a coherent substrate field.

Class IV — Declining Substrate: Outer rim depletion. The SMBH maintains the inner substrate but funding does not extend to the outer disk. A characteristic inner win / outer loss pattern emerges.

Class V-A — Ram Pressure: Asymmetric lambda field from external environmental interaction (galaxy cluster membership).

Class V-B — Substrate Theft: Multi-body SMBH competition. A secondary SMBH or nearby massive galaxy depletes the local substrate field. Confirmed for NGC 7793 via THINGS VLA HI Moment-0 morphology (Walter et al. 2008).

Class VI — Barred Substrate Pipe: The galactic bar channels substrate flux along its axis, creating a dipole injection pattern. Confirmed for NGC 3953 via bar geometry and LINER throttle classification.

3.3 Classification Stability

The six-class boundaries are stable under perturbation of global parameters (lambda, kappa_BH, grid). Sweeping lambda from 0.05 to 0.20 shifts the magnitude of substrate wins but does not reclassify galaxies between classes. This stability indicates physical boundaries in galactic substrate topology rather than model artifacts.

3.4 Coupling Efficiency

The substrate coupling efficiency:

```
eta = c_substrate / v_medium
```

maps directly to the six classes: eta > 1 (Class I, substrate-dominated), eta ~ 1 (Class II, coupled boundary), eta < 1 (Classes III-IV, medium-dominated). The classification describes coupling regimes, not mass distributions.

3.5 Quantitative Comparison: BCM vs Newton vs MOND on Full SPARC 175

A production run was performed on all 175 galaxies in the SPARC sample at grid=256 with 25,000 integration steps per galaxy. Four models are compared:

- **BCM RESCALED:** Four frozen global parameters (lambda=0.1, kappa_BH=2.0, grid=256, layers=8), plus one amplitude coefficient A fit per galaxy via least-squares minimization against the observed rotation curve. The per-galaxy amplitude is a single scalar with no free shape parameters.

- **BCM GLOBAL:** Same global parameters as RESCALED, but with a single amplitude coefficient fit once across the entire 175-galaxy sample (1 parameter total).
- **BCM RAW:** Global parameters only, amplitude = 1.0 (zero fitted parameters). Included as an ablation to verify that the amplitude fit is carrying real information rather than being an arbitrary normalization.
- **MOND:** Standard simple interpolation mu(x) = x/(1+x) with a_0 = 1.2 x 10^-10 m/s^2 (Milgrom 1983). Zero per-galaxy tuning (same M/L as SPARC baryonic decomposition).
- **Newtonian:** Baryonic components only (gas + disk + bulge), zero free parameters.

The reduced chi-squared is computed as:

```
chi^2_red = (1 / (N - p)) * sum( (V_model - V_obs)^2 / errV^2 )
```

where N is the number of radial data points, p is the number of per-galaxy free parameters, and errV is the observational velocity uncertainty (floored at 1 km/s).

Table 1: Summary statistics on the complete 175-galaxy SPARC sample (grid=256, production run, 148.6 min GPU runtime)

Model	Free params	Wins vs Newton	Mean chi2	Median chi2
BCM RESCALED	1 per galaxy	130/175 (74.3%)	63.1	17.7
Newton	0	— (baseline)	157.6	51.6
BCM GLOBAL	1 total	10/175 (5.7%)	870.6	255.0
MOND	0	3/175 (1.7%)	1122.6	343.9
BCM RAW	0	0/175 (0%)	5.0e10	6.9e8

BCM RESCALED achieves the lowest mean reduced chi-squared of any model tested, beating Newton on 74.3% of galaxies and MOND on 98.3%. The global amplitude fit from BCM RESCALED settles at A_global = 1.299 x 10^-4 (dimensional conversion from substrate code units to observed km/s).

Parameter count honesty. BCM RESCALED uses one free parameter per galaxy. BCM GLOBAL (1 parameter total) performs 13.8x worse than RESCALED, and BCM RAW (0 parameters) fails catastrophically. This establishes that the per-galaxy amplitude is doing real physical work, not absorbing model freedom. For comparison, NFW dark matter halo fits employ 2 parameters per galaxy (concentration c and virial mass M_vir), and MOND employs 1 global parameter (a_0). BCM RESCALED sits between NFW and MOND in parameter count.

Bracket-stratified performance:

Bracket	N	Newton chi2	MOND chi2	BCM chi2	BCM wins
Dwarf (V<50)	22	30.3	613.5	**7.5**	12/22 (55%)
Low (50-100)	64	128.3	1086.7	**31.7**	56/64 (88%)
Mid (100-150)	33	218.3	1427.7	**40.4**	27/33 (82%)
High (150-200)	18	84.4	714.3	**57.4**	12/18 (67%)
Massive (V>200)	38	262.7	1406.2	**170.7**	23/38 (61%)

BCM is strongest in the low and mid velocity brackets (88% and 82% wins). The dwarf bracket is competitive at 55% wins, though absolute chi-squared values are small for all models at that scale. The massive bracket is the framework's observed weakness, with 61% wins and a mean chi-squared of 170.7 — higher than any other bracket. This pattern is consistent with the Tully-Fisher analysis in Section 3.6.

Representative individual performance. Among the galaxies where BCM RESCALED produces the largest improvement over Newton:

Galaxy	V_max	Newton chi2	BCM chi2	Improvement
UGC00128	134	2485.3	302.1	8.2x
UGC02487	383	1349.7	174.5	7.7x
UGC00634	108	1172.6	45.4	25.8x
NGC5055	206	1592.0	560.0	2.8x
UGC05716	75	998.8	49.4	20.2x
DDO154	48	486.4	6.7	72.5x
UGC11820	85	559.8	9.1	61.5x

Galaxies where Newton wins and BCM fails are concentrated in the massive bracket: UGC05253 (V=248), UGC03546 (V=262), NGC0801 (V=238), NGC2955 (V=276), UGC02916 (V=218). In these systems, the Newtonian baryonic decomposition alone fits the rotation curve tightly, and BCM over-corrects via the substrate contribution.

3.6 Tully-Fisher Exponent Stability

To test whether the BCM framework reproduces the Tully-Fisher relation, we fit a linear regression on the full sample:

```
log(sigma_energy_integral) = alpha * log(V_max_obs) + intercept
```

where sigma_energy_integral is the integrated substrate energy over the galaxy's rotation curve, and V_max_obs is the observed peak rotation velocity. The fit was performed both on the full sample and on bracket-stratified subsamples, with bootstrap resampling (1000 iterations, seed=42) to produce 95% confidence intervals.

Full-sample fit: alpha = 3.42 (95% CI [3.03, 3.88]), Pearson r = 0.79, r^2 = 0.62. The exponent falls within the classical baryonic Tully-Fisher range [3.0, 4.0] (McGaugh et al. 2000).

Bracket-stratified fits:

Bracket	N	alpha median	95% CI	Pearson r
Dwarf	22	1.06	[-0.63, +3.03]	0.25
Low	64	2.06	[+0.21, +3.98]	0.24
Mid	33	7.44	[+0.66, +13.87]	0.40
High	18	6.20	[-3.43, +16.20]	0.27
Massive	38	**-0.29**	**[-2.36, +2.76]**	-0.03

The massive bracket is the only bracket whose 95% confidence interval falls fully below the classical Tully-Fisher range. The near-zero Pearson r in the massive bracket (r = -0.03) indicates the log(sigma_energy) - log(V_max) correlation has essentially vanished at this scale: the BCM framework's substrate-velocity coupling weakens systematically at V_max > 200 km/s.

The bracket-stratified alpha values span 7.7 units between the mid-bracket maximum (+7.44) and the massive-bracket minimum (-0.29), with an interior peak at the mid velocity range. This scale-dependent profile is an empirical finding from the data and is formally documented in the Test 4 JSON artifact (BCM_v26_Tully_Fisher_Exponent_Stability_20260421_081146).

Interpretation for Paper A. We report alpha = 3.42 as the full-sample Tully-Fisher exponent, but emphasize that the relation is scale-dependent across the BCM framework: the relation holds (within classical range) in the ensemble but breaks at the massive bracket. This is an honest statistical description of the data and does not claim that BCM produces a universal Tully-Fisher exponent across all galactic scales.

3.7 Scale-Dependent Substrate Coupling

The pattern of alpha variation across velocity brackets — small at low v, peaking at mid v, vanishing at massive v — combined with the chi-squared degradation at the massive bracket suggests a scale-dependent transition in the substrate coupling. Whether this transition is numerical (a limitation of the BCM amplitude fit at high v) or physical (a genuine phase transition in the substrate response at high baryonic density) is beyond the scope of Paper A.

The empirical observation stands: the BCM framework as formulated fits rotation curves across 74% of the SPARC sample, dominates the low-to-mid velocity regime, and shows systematic weakness at v_max > 200 km/s. A theoretical extension of the framework addressing this scale-dependence through a modulation of the classical mass-energy term is under development and will be presented in a companion paper (Paper B, in preparation). Paper A's claims do not depend on Paper B.

3.8 Anchor Partition Regime Map (v7)

The Anchor Partition Ratio. We introduce the Anchor Partition Ratio (APR), a direct decomposition of observed SPARC rotation curves that requires no BCM solver execution:

```
APR(r) = max(V_obs²(r) - V_newton²(r), 0) / V_obs²(r)

where V_newton²(r) = V_gas²(r) + V_disk²(r) + V_bul²(r)
```

APR measures what fraction of the observed rotation velocity cannot be explained by visible baryons. It is a property of the SPARC observational data, not of any model. APR_max denotes the maximum APR across the rotation curve; APR_outer denotes the mean APR over the outer third of the disk (r >= 0.67 r_max).

Across 175 galaxies: APR_max median = 0.693; 152/175 (86.9%) have APR_max >= 0.30, meaning the substrate-side term must be substantial in the large majority of SPARC galaxies for the Anchor Equation to hold.

Six-regime structure. Applying APR across the full SPARC175 sample with the regime assignment rules below identifies six empirically distinct substrate coupling regimes:

```
MASS_FLOOR (23 galaxies):          APR_max < 0.30
DWARF_INTERMEDIATE (18 galaxies):  v_max < 50 km/s
SUBSTRATE_PLATEAU (62 galaxies):   50 ≤ v_max < 150, APR_outer ≥ 0.65
MIXED_TRANSITION (29 galaxies):    50 ≤ v_max < 150, APR_outer < 0.65
SUPPRESSION_VALLEY (37 galaxies):  150 ≤ v_max < 300 km/s
ROOT_REENTRY (6 galaxies):         v_max ≥ 300 km/s
```

The SUBSTRATE_PLATEAU regime (APR_outer median 0.773) coincides with the BCM performance peak: BCM wins 91.9% of galaxies in this regime. The SUPPRESSION_VALLEY (APR_outer median 0.471) coincides with the regime where Newton dominates: Newton wins 75.7%. This retrodicts Paper A's bracket performance pattern from a completely independent decomposition, not from the solver.

125 km/s snap. The APR_outer mean drops sharply at 125 km/s:

```
Below 125 km/s: APR_outer_mean = 0.6179 (n=109)
Above 125 km/s: APR_outer_mean = 0.4248 (n=66)
Delta:          +0.1931
```

This boundary aligns with the Tully-Fisher exponent peak (alpha = 7.44) in the mid-velocity bracket (Section 3.6) — two independent measurements identifying the same substrate transition. The snap is a property of the SPARC observational data, not a solver artifact.

ROOT_REENTRY anatomy. The six ROOT_REENTRY galaxies — ESO563-G021, NGC2841, NGC5985, UGC02487, UGC02885, UGC02953 — carry APR_outer 0.51-0.76, indicating substantial outer-disk substrate signal. However, the current BCM solver (Paper A architecture) underperforms Newton on these systems: mean BCM outer RMS = 94.0 versus Newton outer RMS = 78.7. Anatomical analysis confirms the substrate signal is outer-disk dominated in 5/6 systems, and the solver failure is largest in the outer zone. This gap is not evidence that the BCM framework is incorrect; it identifies where the current solver architecture lacks an appropriate operator for ROOT-scale substrate geometry.

One exception: UGC02487 (v_max = 383 km/s), the most extreme ROOT_REENTRY system, is the only BCM win in this regime (BCM RMS 92.4 versus Newton RMS 109.5). This system is the primary calibration target for ROOT_REENTRY operator development.

3.9 Macro-Torsion Operator — Proposed Extension (v7)

To address the ROOT_REENTRY underfit, we propose a Macro-Torsion operator:

```
T_μν(r) = η · Θ(v_max - v_crit) · |∂v_ϕ/∂r - v_ϕ/r| · ê_ϕ
```

where: η = coupling viscosity coefficient (free parameter) Θ = Heaviside step function at v_crit = 300 km/s |shear| = |∂v_ϕ/∂r - v_ϕ/r| = magnitude of differential rotation

The shear term |∂v_ϕ/∂r - v_ϕ/r| is computable directly from observed rotation curves without solver execution. For ROOT_REENTRY galaxies with flat or declining outer rotation, ∂v_ϕ/∂r ≈ 0 and v_ϕ/r > 0, producing a negative shear that characterizes their differential rotation structure. The absolute value ensures the torsion tensor responds to the magnitude of differential rotation rather than its sign.

Regime isolation. Applied across all six APR regimes, the Heaviside gate produces strict structural isolation: η sensitivity = 53% for ROOT_REENTRY (v_max > 300 km/s, Θ=1) and exactly 0.000% for all other regimes (Θ=0). The isolation is architectural; η can take any value and produces zero effect on any galaxy below v_crit.

Performance. On the six ROOT_REENTRY galaxies at test parameter η = 100, the operator reduces mean outer RMS from 113.5 (baseline without solver) to 58.4 — a 48% reduction. The complete η sweep:

```
η = 0:    outer RMS = 124.8
η = 10:   outer RMS = 116.8
η = 50:   outer RMS = 88.4
η = 100:  outer RMS = 58.4
```

Calibration status. η = 100 is a first probe parameter, not a physically calibrated value. v_crit = 300 km/s matches the ROOT_REENTRY regime boundary and is well motivated. Physical calibration of η requires outer-disk kinematic data for ROOT_REENTRY galaxies at ALMA resolution. The operator is not yet integrated into the BCM solver. Paper A's numerical results (Section 3.5) do not incorporate this operator. It is presented here as a falsifiable direction: if correct, the six ROOT_REENTRY galaxies should show improved residuals when the solver is extended with this term.

Volume Dilatant — rejected as formulated. A companion operator (Volume Dilatant, Λ_vol) was tested but rejected: it worsened ROOT_REENTRY outer RMS and damaged control regime performance (SUPPRESSION_VALLEY degraded by 305%) due to per-galaxy normalization that removed all mass discrimination. A physically anchored reformulation using absolute mass scaling (M_encl/M_0, M_0 = 10^{11}

M_sun) may be viable as a future extension; the current formulation is not confirmed.

4. Binary Stellar Dynamics

4.1 Tidal Hamiltonian

The binary substrate bridge system places two stellar pumps on a shared wave grid. The tidal Hamiltonian:

```
H_tidal(m) = (v_A + v_tidal - Omega*R_tach/m)^2
```

where v_tidal = Omega_orb * R_tach / 2 is derived from the orbital period with no free parameters. This resolved the first stellar Class VI: HR 1099 observed at m=2 vs standard Alfven prediction m=12 — the tidal correction produces the correct mode.

4.2 Three Binary Systems

System	Mass ratio	Separation	Synchronization	Key result
Alpha Centauri	3.5:1	23.4 AU	No	Coherent bridge, continuous drain
HR 1099	14:1	~10 R_sun	Yes (P_orb=2.84d)	Resistant to colonization, I_B > 0
Spica	8.4:1	~30 R_sun	No (P_orb=4.014d)	One-way drain confirmed

4.3 Phase-Lock Coherence Law

Amplitude stress testing from 8.4:1 to 420:1 pump ratio reveals that cos(delta_phi) remains +1.000000 through 336:1, with first hairline fracture (0.999999) at 420:1. The Phi_reach metric (flood-fill connectivity of ultra-coherent pixels) reproduces four independently derived metrics (sig_drift, I_B, TCF rate, Psi_Phi) in the same ordering without additional parameters.

Burdick Phase-Lock Coherence Law: Synchronization maximizes coherent reach; coherent reach determines flow regime; flow regime determines survival.

4.4 External Validation

HR 1099 shows no mass transfer for 70-80 Myr (Fekel 1983), consistent with the I_B > 0 resistant prediction. Orbital period oscillates rather than decaying (Donati et al. 1999), consistent with alternating time-cost. Spica shows Struve-Sahade effect confirming one-way drain spectroscopically.

5. Falsifiable Predictions

5.1 Neutrino Maintenance Luminosity and Flavor Gradient

The substrate maintenance cost lambda produces a quantitative prediction for each galaxy's neutrino luminosity. Back-calculating the substrate field sigma(r) from the BCM rotation curve fits (Section 3.5) and computing the energy required to maintain that field against lambda decay yields a maintenance luminosity L_nu for each galaxy.

The maintenance luminosity scales as:

```
L_nu ~ v_max^3.9
```

independently recovering the baryonic Tully-Fisher relation (L ~ v^4; Tully & Fisher 1977) from the same four frozen global parameters used for rotation curve fitting. This scaling was not an input to the model — it emerges from the substrate energy budget.

For the 29-galaxy sample, maintenance luminosities range from 10^36 erg/s (dwarf galaxies) to 10^43 erg/s (massive galaxies). All predicted luminosities remain sub-Eddington (L_nu/L_Edd < 6 x 10^-4), confirming physical self-consistency: no galaxy requires more neutrino power than its SMBH can provide.

The predicted neutrino energy flux at Earth exceeds the IceCube point-source sensitivity (10^-12 GeV cm^-2 s^-1) by 2-5 orders of magnitude for all galaxies within 20 Mpc, suggesting that the signal may already be present in existing data as unrecognized extended emission.

The unique BCM signature is the radial gradient: Class IV galaxies (inner-funded, outer-depleted) should exhibit higher neutrino density at inner radii declining to background at the torus edge. Class I galaxies (fully funded) should show uniform emission across the disk. Testable by angular stacking of IceCube/KM3NeT events around nearby SPARC galaxies. Null test: the gradient should vanish for Class I targets.

5.2 Betelgeuse Recovery

The m=3 substrate pattern was disrupted by the Great Dimming (Montarges et al. 2021). BCM predicts full stable m=3 recovery circa 2030-2031 (~11.5 years post-event = one substrate node settle time). The coupling efficiency eta = 0.98% (convection 102x faster than substrate phase speed) establishes the recovery timescale.

5.3 Binary Mass Transfer

Tidally synchronized binaries will exhibit reduced mass transfer rates compared to unsynchronized binaries at similar separation. Testable against existing RS CVn, Algol-type, and W UMa contact binary catalogs.

5.4 Mercury Magnetosphere

BepiColombo m=1 magnetosphere measurement prediction on record before data arrives.

5.5 Benchtop Falsification — Substrate Mode Selection Device (SMSD)

The Substrate Mode Selection Device (SMSD) is a proposed $225-$230 benchtop apparatus designed to test BCM's discrete substrate mode structure at laboratory scale. It has not yet been built. The design is published here so that students or independent researchers can replicate and falsify the prediction without requiring observational infrastructure.

Physical principle. The SMSD is a Taylor-Couette cell: a rotating inner cylinder inside a stationary outer cylinder, with the annular gap filled with galinstan (room-temperature liquid metal). Four N52 neodymium magnet blocks positioned 90 degrees apart around the outer cylinder create an external magnetic field through the fluid. Three SS49E Hall sensors positioned 120 degrees apart on the outer cylinder measure azimuthal magnetic field variations as the inner cylinder rotates. BCM predicts that the magnetohydrodynamic response of the conducting fluid will organize into discrete azimuthal modes rather than a smooth continuum — the substrate mode staircase.

Bill of Materials:

```
1.  Clear acrylic outer cylinder, ID 80mm, wall 5mm, length 150mm
2.  316 stainless steel inner cylinder, OD 60mm, length 140mm,
    smooth machined surface
3.  Galinstan liquid-metal, fills 10mm annular gap
4.  Top and bottom acrylic end caps
5.  Top and bottom 608ZZ bearings
6.  PTFE seal ring (base)
7.  Viton O-rings (base)
8.  Flexible shaft coupler
```

```
9.  NEMA 17 stepper motor
10. A4988 stepper driver board
11. Arduino Uno R3 controller
12. SS49E Hall sensors, 3 total, 120 degrees apart
13. N52 neodymium magnet blocks, 4 total, 90 degrees apart
14. Magnet mount bracket
15. Gap spacer inserts: 5mm, 10mm, 15mm (Configuration C variants)
16. 12V power supply, wiring, and connectors

Estimated build cost: $225-$230 USD
```

Three test configurations:

Configuration A — Discrete Mode Staircase. Sweep inner cylinder rotation rate (omega_in) from 0 to maximum. BCM predicts the Hall sensor output will show discrete steps in azimuthal mode number as rotation rate increases — the mode staircase. A smooth monotonic increase in signal with no steps would falsify this prediction.

Configuration B — m=1 Lock. At a specific rotation rate, BCM predicts a single dominant azimuthal mode (m=1) will lock and rotate coherently, visible as a stable periodic signal across all three Hall sensors with 120-degree phase offsets. Absence of this locked coherent mode at any rotation rate would falsify BCM's mode-selection mechanism.

Configuration C — Boundary Analog. Replace the 10mm annular gap with 5mm or 15mm gap spacer inserts. BCM predicts the mode structure shifts with gap geometry — specifically that the critical rotation rate for mode locking scales with the gap-to-radius ratio. This tests whether the substrate sigma_crit boundary behavior is geometrically encoded in the fluid response.

Falsification criteria. The SMSD falsifies BCM if: (1) no discrete mode steps appear in Configuration A across the full rotation range; (2) no coherent m=1 lock appears in Configuration B; or (3) mode structure is invariant to gap geometry change in Configuration C. Any one of these three is sufficient for falsification. All three passing is necessary but not sufficient for confirmation — the device tests the mode-selection mechanism, not the full substrate model.

The SMSD is accessible to any physics student or independent researcher. No clean room, no cryogenic infrastructure, and no telescope time required. The Arduino controller and A4988 driver board are standard maker-community components. Galinstan is available from specialty chemical suppliers. The full design was developed by Stephen Justin Burdick Sr. (Emerald Entities LLC — GIBUSH Systems) and is published here under the same CC BY-NC-ND 4.0 license as this paper.

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6. Discussion

6.1 Comparison to Existing Approaches

BCM differs from both dark matter and modified gravity in its ontological claim: the gravitational signal attributed to dark matter is modeled as the maintenance cost of spatial substrate, not missing mass or modified force law. The six-class taxonomy provides a physical classification scheme that neither NFW halo fitting (which requires per-galaxy parameters) nor MOND (which provides a universal acceleration scale but no classification) offers.

The chi-squared comparison (Section 3.5) demonstrates that BCM with one amplitude parameter per galaxy achieves a lower mean chi-squared than MOND (zero per-galaxy parameters) and performs competitively against NFW (two fitted parameters per galaxy, typical in SPARC literature analyses). The BCM amplitude parameter is a single scalar with no shape freedom. This does not prove the substrate model is correct — it demonstrates that the substrate- wave approach is competitive with established dark matter and modified-gravity alternatives on a broad galactic sample.

6.2 Limitations

The framework has significant limitations that must be addressed in future work:

The model has not been tested against gravitational lensing observations. This is the primary validation gap — dark matter alternatives historically encounter difficulties reproducing lensing constraints, and BCM may fail here.

The model has not been tested against CMB power spectrum constraints, large-scale structure formation, or Bullet Cluster dynamics — all areas where Lambda-CDM performs well.

The physical mechanism for substrate maintenance (SMBH neutrino flux) is a proposed interpretation, not a derived result. The model functions phenomenologically regardless of the physical funding mechanism; the chi-squared results (Section 3.5) depend only on the wave equation and its parameters, not on the neutrino interpretation. The substrate field operates on a 2+1 dimensional manifold; the projection rules connecting this to 3+1 dimensional spacetime curvature are the subject of ongoing theoretical work.

The per-galaxy amplitude scaling applied to the substrate velocity profile (Section 2.3) means that BCM tests the shape of the rotation curve contribution, not its absolute magnitude. A future version must derive the amplitude from first principles.

6.3 AI Collaboration Methodology

The framework was developed through directed collaboration with four independent AI systems: Claude (Anthropic) for code execution, ChatGPT (OpenAI) for adversarial audit and kill conditions, Gemini (Google) for engineering formalization, and Grok (xAI) for anomaly detection. Each AI was given a different role and different information. The theoretical direction and all original concepts were provided by the sole human author. This methodology — using AI systems as probes with different sampling functions directed by a human controller — is itself a contribution to the practice of computational physics research.

6.4 Reproducibility

All code is open source (GitHub: Joy4joy4all/Burdick-Crag-Mass). All results are published on Zenodo with timestamped JSON evidence across 24 versions (base DOI: 10.5281/zenodo.19251192). Every covariance requirement was evaluated and documented. The framework is fully reproducible.

6.5 Einstein Coupling and Null Pump (v23)

The substrate stress-energy tensor $T_{\mu\nu}$ was derived explicitly from the physical velocity excess via compare_rotation(). The Newtonian limit holds for 4/5 galaxies (M_sub from 9.27×10■ to 1.26×10^{12} M■). Conservation $\nabla_\mu T^{\mu\nu}$ at machine precision (10■²■) for all 5. NGC3953 (Class VI barred) fails the Newtonian limit — Newton already wins. This failure is the covariance requirement: the gate must be able to say NO.

The null pump test (J=0) proved substrate collapse: $\sigma \to 0.000000$ in all 5 galaxies. Rotation fits degrade by +129 to +188 km/s. PUMP ESSENTIAL: 5/5. The substrate is a funded state.

6.6 Boundary Dynamics and σ_{crit} Discovery (v24)

Three substrate perturbation regimes were identified using synthetic binary source fields:

Diffusive Healing: stochastic perforation of J (±40% noise) heals completely (coherence 0.99987). The solver self-averages neutrino injection noise.

Coherence Failure: baryonic density noise degrades coherence to 0.742 at the binary throat. The mechanism is local maintenance cost increase, not injection pattern disruption.

Boundary Nonlinear: perturbation at 80% r_max produces 38× σ overshoot. The torus edge dissolves into bulk without a nonlinear saturation term. A boundary stability sweep across 8 configurations showed that only a σ_crit clamp maintains a stable thin edge. Dissipation alone is insufficient. Injection is catastrophic.

The dimensionless control parameter Π = σ_edge/σ_crit governs boundary stability. Alpha Centauri (Π << 1): stable at all approach angles. HR 1099 (Π > 1): requires active clamping. The safe arrival corridor is defined by min(|∇σ|) — minimum gradient magnitude, not absolute σ level.

This identifies a nonlinear saturation mechanism as a required component of the substrate field equation at boundaries. Whether σ_crit derives from system parameters (separation, pump ratio, λ) or must be independently measured is the subject of ongoing work.

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7. Conclusions

The Burdick Crag Mass framework demonstrates that galactic rotation curves can be reproduced without dark matter or modified gravity by treating the gravitational signal as a substrate maintenance cost. On the complete 175-galaxy SPARC sample, BCM with four frozen global constants plus one amplitude coefficient per galaxy beats Newtonian baryonic fits on 130 of 175 galaxies (74.3%) and beats MOND on 172 of 175 (98.3%), achieving a mean reduced chi-squared of 63.1 compared to Newton's 157.6 and MOND's 1122.6. An ablation study confirms the per-galaxy amplitude carries real physical information: a single global amplitude fit performs 13.8x worse, and an unscaled fit fails catastrophically. The six-class taxonomy emerging from the framework classifies galaxies into physically distinct substrate interaction states that are stable under parameter perturbation.

The Tully-Fisher exponent on the full sample is alpha = 3.42 (95% CI [3.03, 3.88]), consistent with classical baryonic TF. Bracket-stratified bootstrap reveals substantial scale-dependence across the framework: the TF relation holds (within classical range) in the ensemble but breaks down at the massive velocity bracket (alpha = -0.29, 95% CI fully below classical). This empirical observation points toward a scale-dependent transition in the substrate coupling that warrants theoretical extension. Binary stellar dynamics on three systems reproduce observed mass transfer behavior.

The null pump test (v23) proved the substrate is a funded state: σ → 0 without SMBH injection. The Einstein coupling derives T_μν at machine-precision conservation with substrate enclosed masses spanning five orders of magnitude.

The boundary dynamics tests (v24) identified three distinct perturbation regimes and discovered that a nonlinear saturation limit (σ_crit) is required for stable boundary formation. This provides a falsifiable prediction: the torus edge of any galaxy should exhibit a measurable σ_crit that correlates with system parameters.

A theoretical extension of the framework addressing the scale-dependent substrate coupling observed in the Tully-Fisher bracket analysis is under development and will be presented in a companion paper. Paper A's claims do not depend on that extension.

v7 additions (Anchor Partition Regime Map). The v7 additions establish that the scale-dependent behavior observed in the TF bracket analysis corresponds to an identifiable structure in the observational data itself. The six-regime APR map shows: (1) substrate coupling peaks at 50-150 km/s where BCM wins most; (2) APR drops sharply at 125 km/s, aligning with the TF exponent peak; (3) ROOT_REENTRY galaxies (v_max > 300 km/s) carry real substrate signal in their outer disks but the current solver architecture lacks the appropriate operator. The Macro-Torsion operator proposed in Section 3.9 provides a regime-safe directional extension; its integration into the solver and physical calibration are reserved for a future version. The ROOT_REENTRY finding generates a specific falsifiable prediction: the six galaxies in that regime should show outer-disk rotation residuals consistent with differential rotation shear coupling when observed at sufficient spatial resolution.

The framework generates multiple falsifiable predictions testable with existing observational data and a $225 benchtop device.

Whether or not the substrate model ultimately describes physical reality, the computational methodology — adversarial multi-AI collaboration with full reproducibility — provides a template for future theoretical physics research.

Space is not a container. Space is a maintenance cost.

Acknowledgments

All theoretical concepts originated with Stephen Justin Burdick Sr. Code execution: Claude (Anthropic). Adversarial computation: ChatGPT (OpenAI). Engineering formalization: Gemini (Google). Anomaly detection: Grok (xAI). Named in honor of Bruce Burdick (Brucetron discovery). The SPARC dataset was provided by Lelli, McGaugh, and Schombert. THINGS VLA HI data from Walter et al. GW150914 waveform template from LIGO Open Science Center.

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Stephen Justin Burdick Sr. — Emerald Entities LLC GIBUSH Systems — 2026 All code and data: https://github.com/Joy4joy4all/Burdick-Crag-Mass Zenodo v6: 10.5281/zenodo.19680280 | *Base: 10.5281/zenodo.19251192*